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Q1. Write a C program to find the maximum and minimum elements in an array.

```c
#include<stdio.h>
int main(){
    int n,min,max;

    printf("Enter the size of array:");
    scanf("%d",&n);

    int arr[n];
    printf("Enter the elements of array:");

    for(int i=0;i<n;i++) scanf("%d",&arr[i]);
    min=max=arr[0];

    for(int i=1;i<n;i++)
    {
        if(arr[i]<min)min=arr[i];
        if(arr[i]>max)max=arr[i];
    }

    printf("Minimum:%d\n",min);
    printf("Maximum:%d\n",max);
    return 0;
}
```

Output:

```
Enter the size of array:3
Enter the elements of array:1
9
8
Minimum:1
Maximum:9
```

Q2. Implement a C program to reverse the elements of an array.

```c
#include<stdio.h>
int main(){
    int n,temp;
    printf("Enter the size of array:");
    scanf("%d",&n);

    int arr[n];
```

```
    printf("Enter the elements of array:");
    for(int i=0;i<n;i++) scanf("%d",&arr[i]);

    printf("The elements of the array are:");

    for(int i=0;i<n;i++) printf("%d",arr[i]);
    printf("\n");
    for(int i=0;i<n/2;i++)
    {
        temp=arr[i];
        arr[i]=arr[n-1-i];
        arr[n-1-i]=temp;
    }
    printf("The elements of array after reversing are:");

    for(int i=0;i<n;i++) printf("%d",arr[i]);
    printf("\n");
    return 0;
}
```

Output:

```
Enter the size of array:5
Enter the elements of array:3
4
8
6
7
The elements of the array are:34867
The elements of array after reversing are:76843
```

Q3. Write a C program to sort an array of integers in ascending order using a sorting algorithm of your choice (e.g., bubble sort, selection sort, insertion sort).

```
#include<stdio.h>
int main(){
    int n,temp;
    printf("Enter the size of array:");
    scanf("%d",&n);

    int arr[n];
    printf("Enter the elements of array:");
    for(int i=0;i<n;i++) scanf("%d",&arr[i]);

    printf("The elements of the array are:");
    for(int i=0;i<n;i++) printf("%d",arr[i]);
```

```
    printf("\n");

    for(int i=0;i<n-1;i++)
    {
        for(int j=0;j<n-1-i;j++)
        {
            if(arr[j]>arr[j+1])
            {
                temp=arr[j];
                arr[j]=arr[j+1];
                arr[j+1]=temp;
            }
        }
    }
    printf("The elements of array after sorting are:");
    for(int i=0;i<n;i++) printf("%d",arr[i]);
    printf("\n");

    return 0;
}
```

Output:

```
Enter the size of array:3
Enter the elements of array:5
9
7
The elements of the array are:597
The elements of array after sorting are:579
```

Q4. Implement a C program to find the second largest element in an array.

```
#include <stdio.h>

int main() {
    int arr[] = {12, 35, 1, 10, 34, 1};
    int n = sizeof(arr) / sizeof(arr[0]);

    int largest, second;


    if (arr[0] > arr[1]) {
        largest = arr[0];
        second = arr[1];
    } else {
        largest = arr[1];
```

```
        second = arr[0];
    }


    for (int i = 2; i < n; i++) {
        if (arr[i] > largest) {
            second = largest;
            largest = arr[i];
        } else if (arr[i] > second && arr[i] != largest) {
            second = arr[i];
        }
    }

    printf("Second Largest Element: %d\n", second);

    return 0;
}
```

Output:

```
Second Largest Element: 34
```

Q5. Write a C program to merge two sorted arrays into a single sorted array.

```
#include<stdio.h>
int main(){
    int n1,n2;

    printf("Enter the size of the first array:");
    scanf(" %d",&n1);

    printf("Enter the size of the second array:");
    scanf(" %d",&n2);

    int arr1[n1], arr2[n2];

    printf("Enter the value of first array:");
    for (int i=0; i<n1; i++)
    scanf(" %d", &arr1[i]);

    printf("Enter the value of second array:");
    for (int j=0; j<n2; j++)
    scanf(" %d", &arr2[j]);

    int x=0, y=0, z=0;

    int merged_arr[n1+n2];
```

```
    while(x<n1 || y<n2)
    {
        if (arr1[x]<arr2[y]) merged_arr[z++]=arr1[x++];
            else
            merged_arr[z++]=arr2[y++];

    }
    printf("Merged array:");
    for (int k=0; k<n1+n2; k++)
    printf(" %d",merged_arr[k]);
    printf("\n");

    return 0;
}
```

Output:

```
Enter size of first and second array: 2
3
Enter value for first array: 2
4
Enter value for second array: 9
8
5
Merged array: 2 4 9 8 5
```

Q6. Write a C program to find the length of a string without using the built-in string functions.

```
#include<stdio.h>
int main(){
    int count=0;
    char abc[100];

    printf("Enter a string:");
    scanf(" %s", abc);

    for (int i=0; i<=100; i++)
    {
        if (abc[i]=='\0')
        {
            break;
        }
        count++;
    }
    printf("The length of the string is: %d", count);
```

```
    return 0;
}
```

Output:

```
Enter a string:Apple
The length of the string is: 5
```

Q7. Implement a C program to reverse a string.

```
#include <stdio.h>

int main() {
    char str[100];
    int i, length = 0;
    char temp;

    printf("Enter a string: ");
    scanf("%s", str);

    while (str[length] != '\0') {
        length++;
    }

    for (i = 0; i < length / 2; i++) {
        temp = str[i];
        str[i] = str[length - 1 - i];
        str[length - 1 - i] = temp;
    }

    printf("Reversed string: %s\n", str);

    return 0;
}
```

Output:

```
Enter a string: fire
Reversed string: erif
```

Q8. Write a C program to check if a given string is a palindrome.

```
#include<stdio.h>
#include<string.h>
#include<ctype.h>
int main(){
    char str[100];
```

```
    int i , length;
    int isPalindrome=1;

    printf("Enter a string:");
    scanf("%s", str);

    for(i=0; str[i]; i++){
        str[i]= tolower(str[i]);
    }

    length=strlen(str);

    for(i=0; i<length/2; i++)
    {

        if(str[i] != str[length-i-1])
        {
            isPalindrome=0;
            break;
        }
    }
    if(isPalindrome)
    {
        printf("%s is a palindrome",str);
    }
    else{
        printf("%s is not a palindrome",str);
    }
    return 0;
}
```

Output:

```
Enter a string:Radar
radar is a palindrome
```

Q9. Implement a C program to count the occurrence of a specific character in a string.

```
#include <stdio.h>
#include<ctype.h>
int main(){

    char str[100];
    int i, count=0;
    char ch;

    printf("Enter a string:");
    scanf(" %s", str);
```

```
    printf("Enter the character to count:");
    scanf(" %c", &ch);

    for (i=0; str[i]!='\0'; i++)
    {
       if (tolower(str[i])==tolower(ch))
        {
            count++;
        }
    }
    printf(" %c comes %d times in the string %s\n", ch, count, str);
    return 0;
}
```

Output:

```
Enter a string:banana
Enter the character to count:A
 A comes 3 times in the string banana
```

Q10. Write a C program to concatenate two strings without using the built-in string functions.

```
#include<stdio.h>
int main(){

    char str1[100];
    char str2[100];

    printf("Enter the first string:");
    scanf(" %s", str1);

    printf("Enter the second string:");
    scanf(" %s", str2);

    int i=0;
    while (str1[i]!= '\0')
    {
        i++;
    }

    str1[i]= ' '; //adding a space
    i++;

    int j=0;
    while (str2[j]!= '\0')
    {
```

```
        str1[i]=str2[j];
        i++;
        j++;
    }

    str1[i]= '\0';
    printf("The concatenated string is %s", str1);


    return 0;


}
```

Output:

```
Enter the first string:C
Enter the second string:Programming
The concatenated string is C Programming
```

Q11. Write a function named calculateAverage that takes an array of integers as input and returns the average of the numbers.

```
#include <stdio.h>
float calculateAverage (float arr[], int size)
{
    int sum=0;
    float average;

    for (int i=0; i<size; i++)
    {
        sum=sum+arr[i];
    }
    average=(float)sum/size;
    return average;
}
int main(){
    float arr[100];
    int n;
    float avg;

    printf("Enter the no. of integers:");
    scanf(" %d", &n);

    printf("Enter %d no. of integers\n",n);
    for (int i=0; i<n; i++)
    {
```

```
        printf("Number %d:",i+1);
        scanf(" %f" , &arr[i]);
    }

    avg=calculateAverage (arr,n);
    printf("\nAverage: %.2f\n",avg);

    return 0;
}
```

Output:

```
Enter the no. of integers:5
Enter 5 no. of integers
Number 1:2
Number 2:5
Number 3:4
Number 4:1
Number 5:3

Average: 3.00
```

Q12. Write a function named isPalindrome that takes a string as input and returns 1 if it is a palindrome (reads the same forwards and backwards), and 0 otherwise.

```
#include<stdio.h>
#include<string.h>
#include<ctype.h>
int isPalindrome (char str[])
{
    int length;

    for(int i = 0; str[i] != '\0'; i++)
     {
        str[i] = tolower(str[i]);
     }

    length=strlen(str);

    for (int i=0; i<length/2; i++)
    {
        if (str[i] != str[length-i-1])
        {
            return 0; //not palindrome
        }
    }
```

```
    return 1;
}
int main(){
    char str[100];
    int result;

    printf("Enter a string:");
    scanf(" %s", str);

    result=isPalindrome(str);

    if(result==1){
        printf("Palindrome\n");
    }
    else{
        printf("Not palindrome\n");
    }
    return 0;
}
```

Output:

```
Enter a string:Bob
Palindrome
```

Q13. Write a function named findFactorial that takes an integer as input and returns its factorial.

```
#include<stdio.h>
int findFactorial (int n)
{
    int i=1;
    int factorial=1;

    while(i<=n)
    {
        factorial=factorial*i;
        i++;
    }
    return factorial;
}

int main(){
    int n;

    printf("Enter a number:");
    scanf(" %d", &n);
```

```
    int factorial=findFactorial(n);

    printf(" %d! = %d",n, factorial);
    return 0;
}
```

Output:

```
Enter a number:5
 5! = 120
```

Q14. Write a function named convertTemperature that takes a temperature value in Celsius and converts it to Fahrenheit. The function should return the converted temperature.

```
#include <stdio.h>
int convertTemperature (float celsius)
{
    float fahrenheit;

    fahrenheit=(celsius*9/5)+32;
    return fahrenheit;
}
int main(){
    float celsius, fahrenheit;

    printf("Enter the temperature in celsius:");
    scanf(" %f", &celsius);

    fahrenheit=convertTemperature(celsius);
    printf(" %.2f%cC is %.2f%cF",celsius, 248, fahrenheit, 248);
    return 0;
}
```

Output:

```
Enter the temperature in celsius:0
 0.00°C is 32.00°F
```

Q15. Write a function named countOccurrences that takes a string and a character as input and returns the number of times the character appears in the string.

```
#include<stdio.h>
#include<ctype.h>
int countOccurrences (char str[], char ch)
```

```
{
    int count=0;

    for (int i=0; str[i]; i++)
    {
        str[i]=tolower(str[i]);
    }

    for (int i=0; str[i] != '\0'; i++)
    {
        if(str[i]==ch)
        {
            count++;
        }
    }
    return count;
}
int main(){
    char str[100];
    char ch;
    int result;

    printf("Enter a string:");
    scanf(" %s", str);

    printf("Enter character to count:");
    scanf(" %c", &ch);

    result=countOccurrences (str, ch);
    printf("%c comes %d times in the string %s",ch, result, str);

    return 0;
}
```

Output:

```
Enter a string:banana
Enter character to count:a
a comes 3 times in the string banana
```

Q16. Write a function named reverseArray that takes an array of integers as input and reverses the order of the elements in the array.

```
#include<stdio.h>
```

```c
int reverseArray (int arr[],int size)
{
    for ( int i=0; i<size/2; i++)
    {
        int temp=arr[i];
        arr[i]=arr[size-i-1];
        arr[size-i-1]=temp;
    }
    return 1;
}
int main(){
    int arr[100];
    int n, result;

    printf("Enter the no. of integers:");
    scanf(" %d", &n);

    printf("Enter the integers:");
    for (int i=0; i<n; i++)
    {
        scanf(" %d", &arr[i]);
    }
    result=reverseArray (arr,n);

    printf("After reversing:");
    for (int i=0; i<n; i++)
    {
        printf(" %d", arr[i]);
    }

    return 0;
}
```

Output:

```
Enter the no. of integers:3
Enter the integers:2
4
9
After reversing: 9 4 2
```

Q17. Write a function named calculatePower that takes two integers, base and exponent, as input and returns the result of raising the base to the exponent.

```c
#include<stdio.h>
int calculatePower(int base, int exponent)
```

```
{
    int result=1;

    for( int i=0; i<exponent; i++)
    {
        result=result*base;
    }
    return result;
}

int main(){
    int base, exponent;

    printf("Enter the base:");
    scanf(" %d",&base);

    printf("Enter the exponent:");
    scanf(" %d",&exponent);

    int result=calculatePower(base,exponent);

    printf("The power is %d\n",result);

    return 0;
}
```

Output:

```
Enter the base:5
Enter the exponent:2
The power is 25
```

Q18. Write a function named findPrimeNumbers that takes an integer n as input and prints all prime numbers from 1 to n.

```
#include <stdio.h>
int findPrimeNumbers (int n)
{
    int isPrime;
    int count=0;

    printf("Prime Numbers from 1 to %d\n",n);

    for (int i=2; i<=n; i++)
    {
        isPrime=1;
        for (int j=2; j<i; j++)
```

```
        {
            if(i%j==0)
            {
                isPrime=0;
                break;
            }
        }
        if (isPrime==1)
        {
            printf(" %d", i);
            count++;
        }
    }
    return count;
}
int main()
{
    int n;
    int Prime;

    printf("Enter a number:");
    scanf(" %d", &n);

    Prime=findPrimeNumbers(n);


    printf("\nTotal Prime Numbers: %d\n", Prime);
    return 0;
}
```

Output:

```
Enter a number:10
Prime Numbers from 1 to 10
 2 3 5 7
Total Prime Numbers: 4
```

Q19. Write a function named calculateFactorialSeries that takes an integer n as input and prints the factorial series up to n.

```
#include <stdio.h>
int calculateFactorialSeries (int n)
{
    int factorial;

    for (int i=1; i<=n; i++)
    {
```

```
        factorial=1;
        for (int j=1; j<=i; j++)
        {
            factorial=factorial*j;
        }
        printf("%d! = %d\n",i,factorial);
    }
    return 1;
}
int main()
{
    int n;
    printf("Enter a number for calculating Factorial Series:");
    scanf(" %d", &n);

    calculateFactorialSeries(n);
    return 0;
}
```

Output:

```
Enter a number for calculating Factorial Series:4
1! = 1
2! = 2
3! = 6
4! = 24
```

Q20. Write a function named calculateGCD that takes two integers as input and returns their greatest common divisor (GCD).

```
#include<stdio.h>
int calculateGCD (int num1, int num2)
{
    int temp;

    while(num2!=0)
    {
        temp=num2;
        num2=num1%num2;
        num1=temp;
    }
    return num1;
}
int main()
{
    int num1,num2;
```

```
    int result;

    printf("Enter first number:");
    scanf(" %d", &num1);
    printf("Enter second number:");
    scanf(" %d", &num2);

    result=calculateGCD (num1, num2);
    printf("\nGCD of %d and %d is %d",num1, num2, result);

    return 0;
}
```

Output:

```
Enter first number:5
Enter second number:20

GCD of 5 and 20 is 5
```

Q21. Write a recursive function named calculateFactorial that takes an integer n as input and returns its factorial.

```
#include<stdio.h>
int calculateFactorial (int num)
{
    if (num==0 || num==1)
    {
        return 1;
    }
    else{
        return num * calculateFactorial(num-1);
    }
}
int main()
{
    int num, result;

    printf("Enter a number:");
    scanf(" %d", &num);

    result=calculateFactorial(num);

    printf("The factorial is %d\n",result);

    printf("%d! = %d",num, result);
```

```
    return 0;
}
```

Output:

```
Enter a number:4
The factorial is 24
4! = 24
```

Q22. Write a recursive function named calculateFibonacci that takes an integer n as input and returns the nth Fibonacci number. The Fibonacci sequence starts with 0 and 1, and each subsequent number is the sum of the two preceding numbers.

```
#include<stdio.h>
int calculateFibonacci (int num)
{
    if(num==0)
    {
        return 0;
    }
    else if(num==1){
        return 1;
    }
    else{
        return calculateFibonacci(num-1) + calculateFibonacci(num-2);
    }
}
int main()
{
    int num, result;

    printf("Enter a number:");
    scanf(" %d", &num);

    result=calculateFibonacci(num);

    printf("Fibonacci (%d) = %d",num, result);

    return 0;
}
```

Output:

```
Enter a number:10
Fibonacci (10) = 55
```

Q23. Write a recursive function named calculateGCD that takes two integers a and b as input and returns their greatest common divisor (GCD).

```
#include<stdio.h>
int calculateGCD (int a, int b)
{
    if(b==0)
    {
        return a;
    }
    else{
        return calculateGCD(b, a%b);
    }
}
int main()
{
    int a,b,result;

    printf("Enter the first number:");
    scanf(" %d", &a);

    printf("Enter the second number:");
    scanf(" %d", &b);

    result=calculateGCD(a,b);

    printf("The GCD of %d and %d is %d",a,b,result);

    return 0;
}
```

Output:

```
Enter the first number:5
Enter the second number:20
The GCD of 5 and 20 is 5
```

Q24. Write a recursive function named calculatePower that takes two integers base and exponent as input and returns the result of raising the base to the exponent.

```
#include<stdio.h>
int calculatePower(int base, int exponent)
{
    if(exponent==0)
    {
```

```
        return 1;
    }
    else{
        return base * calculatePower(base, exponent-1);
    }
}
int main()
{
    int base, exponent;
    int result;

    printf("Enter the base:");
    scanf(" %d", &base);

    printf("Enter the exponent:");
    scanf(" %d", &exponent);

    result= calculatePower(base, exponent);

    printf("The result is %d\n",result);

    printf("%d^%d = %d", base, exponent, result);

    return 0;
}
```

Output:

```
Enter the base:5
Enter the exponent:2
The result is 25
5^2 = 25
```

Q25. Write a recursive function named calculateSumOfDigits that takes an integer n as input and returns the sum of its digits.

```
#include<stdio.h>
int calculateSumOfDigits(int n)
{
    if(n==0)
    {
        return 0;
    }
    else{
        return (n%10) + calculateSumOfDigits(n/10);
    }
```

```
}
int main(){
    int n;

    printf("Enter a number:");
    scanf(" %d", &n);

    int result=calculateSumOfDigits(n);

    printf("The sum of digits is %d", result);

    return 0;
}
```

Output:

```
Enter a number:23
The sum of digits is 5
```

Q26. Write a recursive function named reverseString that takes a string as input and returns the reversed string.

```
#include<stdio.h>
#include<string.h>
int reverseString (char str[], int i, int j)
{
    if(i>=j){
        return 1;
    }
    char temp= str[i];
    str[i]=str[j];
    str[j]=temp;

    return reverseString(str, i+1, j-1);
}
int main(){
    char str[100];
    int result;

    printf("Enter a string:");
    scanf(" %s", str);

    int length=strlen(str);

    result=reverseString(str, 0, length-1);

    if(result==1)
```

```
    {
        printf("Reversed string is %s", str);
    }
    return 0;
}
```

Output:

```
Enter a string:banana
Reversed string is ananab
```

Q27. Write a recursive function named printTriangle that takes an integer n as input and prints a triangle of asterisks (*) with n rows.

```
#include<stdio.h>
int printTriangle(int n)
{
    if(n==0)
    {
        return 1;
    }
    printTriangle(n-1);

    for (int i=0; i<n; i++)
    {
        printf(" *");
    }
    printf("\n");
    return 1;
}
int main()
{
    int n,result;

    printf("Enter the number of rows:");
    scanf(" %d", &n);

    result=printTriangle(n);
    printf("Triangle:%d\n", result);

    return 0;

}
```

Output:

```
Enter the number of rows:3
 *
 * *
 * * *
Triangle:1
```

Q28. Write a recursive function named calculateBinary that takes an integer n as input and returns its binary representation as a string.

```c
#include<stdio.h>
int calculateBinary(int n)
{
    if(n==0)
    {
        return 1;
    }
    calculateBinary(n/2);
    printf("%d", n%2);

    return 1;
}
int main()
{
    int n;

    printf("Enter a number:");
    scanf(" %d", &n);

    printf("The binary of %d is ", n);

    if(n==0)
    {
        printf("0");
    }
    else{
        calculateBinary(n);
    }
    return 0;
}
```

Output:

```
Enter a number:5
The binary of 5 is 101
```

Q29. Write a recursive function named isPalindrome that takes a string as input and returns 1 if it is a palindrome (reads the same forwards and backwards), and 0 otherwise.

```
#include<stdio.h>
#include<string.h>
#include<ctype.h>
int isPalindrome(char str[], int start, int end)
{
    if(start>=end)
    {
        return 1;
    }
          if (tolower(str[start]) != tolower(str[end]))
    {
        return 0;
    }
    return isPalindrome(str, start+1, end-1);
}
int main()
{
    char str[100];

    printf("Enter a string:");
    scanf(" %s", str);

    int length=strlen(str);


    if(isPalindrome(str, 0, length-1))
    {
        printf("%s is a palindrome",str);
    }
    else{
        printf("%s is not a palindrome",str);
    }
    return 0;
}
```

Output:

```
Enter a string:radar
radar is a palindrome
```

Q30. Write a recursive function named countOccurrences that takes a string and a character as input and returns the number of times the character appears in the string.

```c
#include<stdio.h>
#include<ctype.h>
int countOccurrences(char str[], char ch)
{
    if (str[0]=='\0')
    {
        return 0;
    }
    if (tolower(str[0])==tolower(ch))
    {
        return 1+ countOccurrences(str+1, ch);
    }
    return countOccurrences(str+1, ch);
}
int main()
{
    char str[100];
    char ch;

    printf("Enter a string:");
    scanf(" %s", str);

    printf("Enter a character to count:");
    scanf(" %c", &ch);

    int count=countOccurrences(str,ch);

    printf("%c appears %d times in %s",ch, count, str);

    return 0;

}
```

Output:

```
Enter a string:Banana
Enter a character to count:A
A appears 3 times in Banana
```